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Altered flexion-relaxation responses exist during asymmetric trunk flexion movements among persons with unilateral lower-limb amputation

Brad D. Hendershot [a], Maury A. Nussbaum [a,b,*]

[a] Virginia Tech – Wake Forest School of Biomedical Engineering and Sciences, Virginia Tech, Blacksburg, VA 24061, USA
[b] Industrial and Systems Engineering, Virginia Tech, Blacksburg, VA 24061, USA

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ABSTRACT

Repetitive exposures to altered gait and movement following lower-limb amputation (LLA) have been suggested to contribute to observed alterations in passive tissue properties and neuromuscular control in/surrounding the lumbar spine. These alterations, in turn, may affect the synergy between passive and active tissues during trunk movements. Eight males with unilateral LLA and eight non-amputation controls completed quasi-static trunk flexion–extension movements in seven distinct conditions of rotation in the transverse plane: 0° (sagittally-symmetric), ±15°, ±30°, and ±45° (sagittally-asymmetric). Electromyographic (EMG) activity of the bilateral lumbar erector spinae and lumbar kinematics were simultaneously recorded. Peak lumbar flexion and EMG-off angles were determined, along with the difference ("DIFF") between these two angles and the magnitude of peak normalized EMG activities. Persons with unilateral LLA exhibited altered and asymmetric synergies between active and passive trunk tissues during both sagittally-symmetric and -asymmetric trunk flexion movements. Specifically, decreased and asymmetric passive contributions to trunk movements were compensated with increases in the magnitude and duration of active trunk muscle responses. Such alterations in trunk passive and active neuromuscular responses may result from repetitive exposures to abnormal gait and movement subsequent to LLA, and may increase the risk for LBP in this population.

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1. Introduction

An estimated 623,000 people were living with a major lower-limb amputation (LLA) in the US in 2005, a number that is projected to more than double by 2050 (Ziegler-Graham et al., 2008). Chronic secondary physical conditions and pain are common among persons with LLA (Gailey et al., 2008). Low back pain (LBP) in particular, rated as more bothersome than residual and phantom limb pain (Ehde et al., 2001; Smith et al., 1999), represents a substantial secondary impairment among persons with LLA. Due to the coupling of joints in the musculoskeletal system (Dananberg, 1993), alterations in gait and movement patterns among persons with LLA may influence the loading in tissues in/surrounding the spine and thereby contribute to chronic alterations in passive spinal tissues and neuromuscular control. Further, experimental studies indicate larger trunk and pelvic motions during gait compared to able-bodied controls (Goujon-Pillet et al., 2008; Jaegers et al., 1995; Morgenroth et al., 2010); these may lead to (or be consequences of) increases in the demands on the trunk musculature to maintain equilibrium and stability of the spine. Though repetitive exposures to altered gait and movement among persons with LLA may play an important role in the development of LBP in this population, it is currently unclear whether adaptations in trunk passive properties and/or neuromuscular control occur as a result of LLA and subsequent use of a prosthetic device.

Preserving an effective synergy between passive spinal structures and active trunk neuromuscular responses is important for maintaining spinal stability and controlling spinal loads (Panjabi, 1992). As such, differences in the mechanical properties of spinal tissues can alter active-passive load sharing in the spine and surrounding musculature. Reductions in the passive resistance to forward trunk flexion movements, for example, have been related to spinal instability (Graham and Brown, 2012; Olson et al., 2004; Solomonow et al., 2000). Prolonged static trunk flexion reduces passive support of the spine (Bazrgari et al., 2011; Hendershot et al., 2011; McGill and Brown, 1992), delays the onset of muscle relaxation during flexion movements (Shin and Mirka, 2007), and increases muscle activation during flexion/extension movements (Olson et al., 2004). Repeated exposures to abnormal movement

* Corresponding author. Address: Department of Industrial and Systems Engineering, Virginia Tech, 250 Durham Hall (0118), Blacksburg, VA 24061, USA. Tel.: +1 (540) 231 6053; fax: +1 (540) 231 3322.
E-mail address: nussbaum@vt.edu (M.A. Nussbaum).

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mechanics and increased trunk kinematics with LLA could lead to similar alterations in the active and passive contributions to flexion–extension movements.

The myoelectric silence that develops in the posterior trunk musculature as the passive spine resists forward flexion – aka the "flexion-relaxation phenomenon" – has long been used as a means for assessing the noted synergy (e.g., Schultz et al., 1985). The extent of load transfer from active muscles to passive tissues during forward-flexion movements can vary depending on the mechanical properties of these tissues (Adams and Dolan, 1996; Gunning et al., 2001). While reduced and asymmetric intrinsic (passive) trunk mechanical properties have been identified around the neutral spine posture among persons with LLA (Hendershot et al., 2013), it is unclear how these changes in passive properties influence the load-sharing behavior (synergy) between active and passive trunk responses during forward-flexion movements. Therefore, the goal of the present work was to use controlled trunk flexion–extension movements to investigate the effects of LLA on active-passive load sharing mechanisms in the low back. It was hypothesized that persons with LLA would have decreased passive contributions to load sharing compared to non-amputation controls, evidenced by increases in the magnitude and/or duration of trunk muscle activity during forward-flexion movements. We further hypothesized that asymmetries in these behaviors would exist in persons with LLA, as a result of decreased trunk passive properties on the side of amputation.

2. Methods

2.1. Participants

Eight males with unilateral LLA (4 transtibial, 4 transfemoral) and eight male, non-amputation controls completed the study. Mean (SD) age, stature, and body mass for participants with LLA were 41.1 (18.7) years, 175.0 (5.0) cm, and 76.6 (10.2) kg, respectively; corresponding values for the control participants were 36.9 (13.4) years, 174.2 (3.8) cm, and 80.3 (11.4) kg. Participants in the control group were recruited to match participants with LLA, on the individual level, in terms of age ($\leqslant$8 years), stature ($\leqslant$5 cm), and body mass ($\leqslant$5 kg); these were not significantly different ($p > 0.50$) between groups. Reasons for amputation included trauma ($n = 5$), congenital deformity or abnormality ($n = 2$), and cancer ($n = 1$). Participants with LLA were 12.3 (10.1) years post-amputation, wore their prosthetic on a daily basis, and were moderately active without the use of upper-extremity assistive devices. Participants (in both groups) reported being free of any underlying musculoskeletal conditions (excluding amputation) that may confound the results. In particular, none of the participants reported any current or recent (6 months) history of low back pain, or any prior spinal surgeries. These inclusion criteria were the same as our prior studies (Hendershot and Nussbaum, 2013; Hendershot et al., 2013), and in which the same participants were involved. Prior to data collection, each participant completed informed consent procedures approved by the Virginia Tech Institutional Review Board. Participants with LLA wore their prosthetic during all testing procedures.

2.2. Experimental design and procedures

Quasi-static trunk flexion–extension movements were performed in seven distinct conditions of transverse-plane rotation: 0° (sagittally symmetric), ±15°, ±30°, and ±45° (sagitally-asymmetric). The latter three were performed bilaterally, directed to the left and right (Fig. 1A), and this set of movements is similar to that described earlier by Ning et al. (2011). Participants stood in a rigid metal frame, and movements of the pelvis and lower limbs were minimized with a rigid pelvic restraint (across the anterior superior iliac spines) and adjustable strap around the ankles, respectively (Fig. 1B).

Initially, and while in an upright posture in the frame, participants performed maximum voluntary contractions (MVCs) in trunk flexion, extension, and left/right lateral bending. During MVCs, electromyographic (EMG) activities of the bilateral lumbar (L3) erector spinae (ES), rectus abdominis, and external obliques were recorded using rectangular bipolar Ag/AgCl surface electrodes, and following existing electrode placement protocols (Callaghan and Dunk, 2002; Mirka, 1991). Prior to applying electrodes, the skin was prepared using abrasion and cleaned with alcohol, and inter-electrode impedance was maintained below 10 KΩ. Raw EMG signals were preamplified (×100) near the collection site, bandpass filtered (10–500 Hz) and amplified in hardware (Measurement Systems Inc., Ann Arbor, MI, USA), and then sampled at 1000 Hz. Peak EMG-RMS values were identified, and used subsequently to normalize EMG data and qualitatively monitor the magnitude of abdominal activity (see below).

Twenty-one movements, involving three replications of each of the seven conditions, were then performed, with the presentation order of transverse rotation randomized. During each movement, participants slowly (~5 s) flexed forward toward targets (Fig. 1A) until reaching a relaxed, passive hanging position (~3 s) with minimal muscle activity and arms hanging relaxed, and then slowly (~5 s) returned to an upright standing posture. In the initial flexion phase, participants were instructed to approach but not actually touch the targets. Computer-generated auditory signals ("beeps") were used to maintain pacing during movements. Participants remained in a relaxed, upright posture for 30 s after each movement to minimize the effects of accumulated creep and fatigue (Shin and Mirka, 2007; Solomonow et al., 2003). During these movements, EMGs were collected as during MVCs. In addition, lumbar kinematics were sampled at 100 Hz from two triaxial 6DOF inertial measurement units (IMUs: Xsens Technologies, XM-B-XB3, Enschede, The Netherlands) affixed over the T12 spinous process and the superior aspect of the sacrum (S1) using double-sided tape.

2.3. Data analysis

Lumbar flexion angle was defined as the relative difference between the T12 and S1 sensors in the standing (initial) vs. flexed hanging (final) postures (Newcomer et al., 2000; Ning et al., 2011; Van Herp et al., 2000). As such, a zero value corresponded to lumbar flexion in the relaxed, upright standing (initial) posture. The maximum ("peak") lumbar flexion angle was determined from the initial time point when trunk movement stopped (i.e., trunk velocity = 0), as the peak flexion angle may increase during the passive hang due to creep deformation of posterior trunk tissues. All EMG data were rectified and then smoothed using a fourth order, zero-lag Butterworth low-pass filter with a 5 Hz cutoff (Jin et al., 2012). Processed EMG from the ES were then temporally matched to lumbar flexion angles to determine "EMG-off" angles (Ning et al., 2011; Solomonow et al., 2003), using within-trial references to determine thresholds for EMG timing (Jin et al., 2012). Within each movement, the difference ("DIFF") between peak lumbar flexion angle and EMG-off angle was also calculated (Fig. 2). Peak normalized EMG-RMS values ("peak nEMG") for the bilateral ES (L3) were obtained during the flexion and extension phases of each movement, by dividing the mean EMG-RMS during a 100 ms window (centered on the peak value) in each phase by the peak EMG-RMS identified from the MVCs (1.0 = 100%MVC). Abdominal EMG activities were analyzed qualitatively in real-time to ensure that flexor contributions were <5% of maximal values

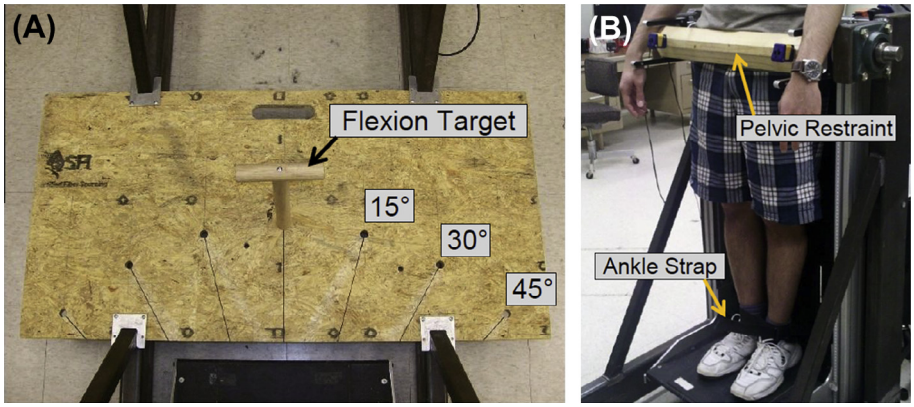

Fig. 1. (A) Top view of the seven distinct conditions of rotation in the transverse plane. The moveable flexion target is depicted in the 0° (sagittally-symmetric) position. (B) Experimental set-up demonstrating the rigid pelvic restraint and ankle strap, respectively used to constrain the pelvis and legs in a rigid frame.

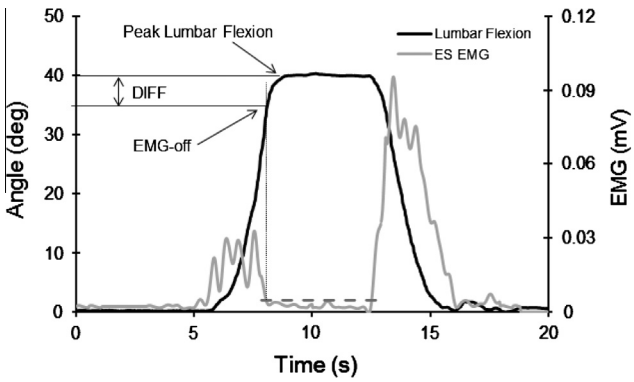

Fig. 2. Representative recordings of the lumbar flexion angle (left axis) and mean bilateral erector spinae (ES) EMG (right axis) from a participant with LLA during a sagittally-symmetric movement. The peak lumbar flexion angle, angle when the ES turned off ("EMG-off"), and the difference ("DIFF") between these are also indicated. The grey horizontal dashed line represents the mean ES EMG + 3SD (during the passive hanging posture) used as a threshold for determining EMG timing.

(from MVCs) during trunk movements and the passive hanging portion of each movement.

2.4. Statistical analyses

Mixed-factor analyses of variance (ANOVA) were used to assess peak lumbar flexion angles, EMG-off angles, DIFF angles, and peak nEMG values between and within groups. One set of ANOVAs was used to compare these responses between groups during sagittally-symmetric (i.e., 0°) movements. A second set of ANOVAs was used to assess the results of sagittally-asymmetric movements (i.e., 15°, 30°, and 45° to the right and left). These asymmetric movements were categorized as being directed to the LEFT and RIGHT for participants in the control group, and ipsilateral (IPSI) and contralateral (CONTRA) within the LLA group. For the latter, movements were defined as being performed away from (CONTRA) or towards (IPSI) the side of amputation. Initial analyses, using mixed-factor ANOVAs, revealed no significant or substantial differences across the three movement replications in each rotation condition. A set of orthogonal *post hoc* contrasts was used to determine: (1) overall differences between groups; (2) bilateral differences among controls; and (3) bilateral differences among participants with LLA. All statistical analyses were performed using JMP (Version 10, SAS Institute, Cary, NC, USA), with statistical significance concluded when $p < 0.05$. Summary statistics are presented as means (SDs). As observed by Ning et al. (2011), and given the low occurrence of muscle relaxation, muscles on the same side as the direction of an asymmetric movement (e.g., left ES during a movement directed to the left) were excluded from these analyses. Because of the low occurrence of muscle relaxation during the 45° transverse rotation condition specifically, EMG-off and DIFF angles during these movements were also excluded from analyses.

3. Results

3.1. Sagittally-symmetric movements

Peak lumbar flexion ($p = 0.26$) and EMG-off ($p = 0.27$) angles were similar between groups, with respective values of 44.1 (4.9)° and 36.3 (4.3)° among participants with LLA and 47.2 (5.3)° and 33.9 (4.0)° among controls. DIFF angles were significantly ($p = 0.02$) larger among controls [13.3 (5.1)°] than participants with LLA [7.9 (3.2)°]. Peak nEMG values were similar between groups during both the flexion ($p = 0.10$) and extension ($p = 0.13$) phases, with respective values of 0.20 (0.11) and 0.51 (0.23) among participants with LLA, and 0.14 (0.04) and 0.39 (0.12) among controls. However, these were larger ($p < 0.0002$) on the intact vs. prosthetic side among participants with LLA in both flexion [0.23 (0.12) vs. 0.17 (0.06)] and extension [0.58 (0.28) vs. 0.44 (0.13)], but bilaterally similar ($p > 0.082$) among controls in flexion [left = 0.13 (0.05) vs. right = 0.15 (0.04)] and extension [left = 0.40 (0.14) vs. right = 0.43 (0.08)].

3.2. Sagittally-asymmetric movements

Peak lumbar flexion angles significantly ($p < 0.001$) decreased with increasing transverse rotation. Specifically, peak lumbar flexion angles were 42.0 (5.3)°, 37.4 (4.3)°, and 33.2 (5.2)° for the 15°, 30°, and 45° transverse rotations, respectively. Peak lumbar flexion angles were also significantly ($p < 0.001$) smaller across all transverse rotations when flexing CONTRA (35.4 (6.8)°) vs. IPSI (38.4 (4.2)°) among participants with LLA, but bilaterally similar ($P = 0.52$) among controls [LEFT = 38.0 (6.4)° vs. RIGHT = 38.3 (6.4)°]. There was a significant ($p = 0.0005$) transverse rotation x movement direction interaction effect; peak lumbar flexion angles were significantly smaller among participants with LLA during CONTRA movements for the 30° ($p < 0.0001$) and 45° ($p < 0.0001$) transverse rotations, but were bilaterally similar (all $p > 0.40$) among controls for all three transverse rotations (Fig. 3).

EMG-off angles also significantly ($p < 0.0001$) decreased with increasing transverse rotation, with respective values of 35.2

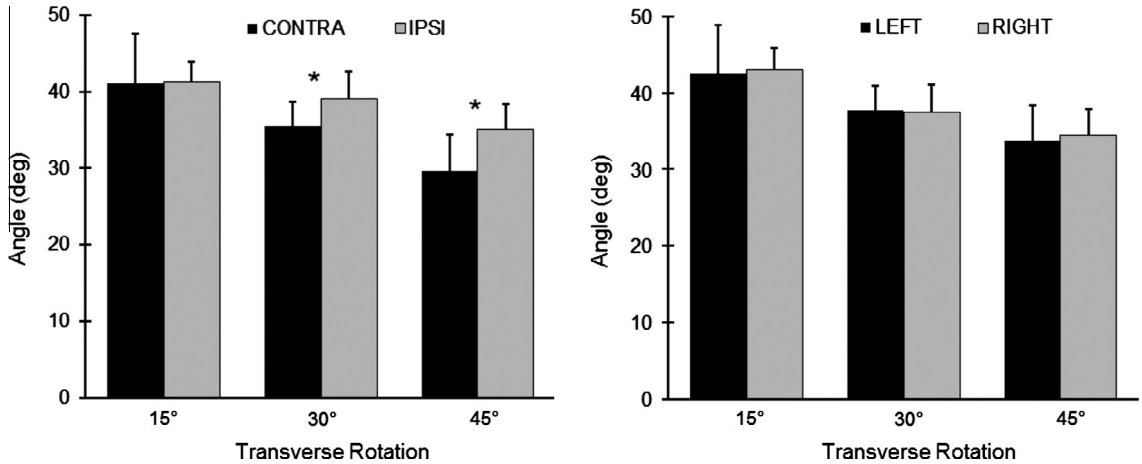

Fig. 3. Peak lumbar flexion angles during sagittally-asymmetric trunk flexion movements among participants with LLA (left) and controls (right). Error bars indicate standard deviations, and asterisks represent significant bilateral differences within a rotation condition.

(4.6)° and 31.3 (3.9)° for the 15° and 30° transverse rotations, but were similar between directions ($p = 0.36$) and groups ($p = 0.93$). DIFF angles were bilaterally similar at both transverse rotation angles among controls ($p = 0.13$), and bilaterally similar at the 15° transverse rotation among participants with LLA ($p = 0.07$), but significantly ($p < 0.001$) smaller during CONTRA movements at the 30° transverse rotation among participants with LLA (Fig. 4).

During flexion, peak nEMG values were similar between groups ($p = 0.16$), transverse rotation angles ($p = 0.72$), and direction ($p = 0.24$). During extension, these were also similar between groups ($p = 0.82$) and directions ($p = 0.42$), but significantly ($p < 0.0001$) decreased with increasing transverse rotation, with respective values of 0.47 (0.26), 0.39 (0.21), and 0.36 (0.28) with rotations of 15°, 30°, and 45°. These responses, however, were more bilaterally asymmetric ($p = 0.012$) among participants with LLA (15.6%) than controls (1.1%), whereby participants with LLA activated their contralateral extensor musculature more when flexing IPSI vs. CONTRA.

4. Discussion

While there were no significant overall group differences, peak lumbar flexion and EMG-off angles tended to be respectively smaller and larger among participants with LLA during sagittally-symmetric movements. There were, though, significantly smaller DIFF angles among participants with LLA. As suggested by Hashemirad et al. (2009), the central nervous system likely monitors the tension developed in posterior spinal tissues and "determines" when the passive tissues can adequately control the remaining portion of the forward-flexion movement without need for active muscle responses. Solomonow et al. (2003) reported a decrease in the DIFF angle (due to increases in the EMG-off angle) following 10 min of static trunk flexion. Smaller DIFF angles among participants with LLA during sagittally-symmetric movements in the present study could therefore indicate similar reductions in passive contributions to forward-flexion movements. If so, additional contributions from active muscles would be required throughout a longer portion of the movement. Further, peak nEMG activities during the flexion and extension phases of each movement were 6–13% MVC larger among participants with LLA compared to controls, supporting the idea that increased muscle activity may be required to overcome reductions in passive contributions to joint stiffness throughout the movements. Increases in trunk muscle activities were found earlier in the same LLA group during a seated balance task, which were similarly suggested to compensate for reduced passive contributions to joint stiffness (Hendershot and Nussbaum, 2013).

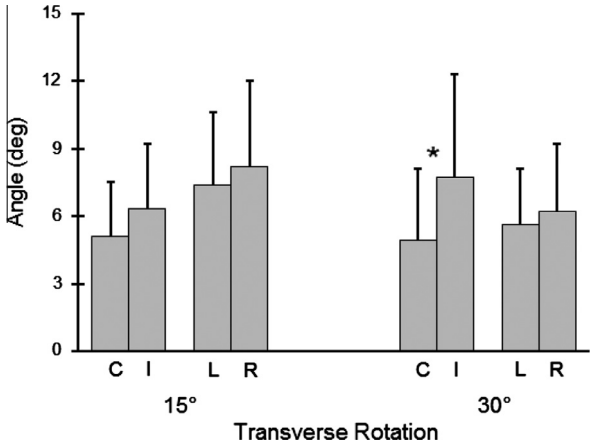

Fig. 4. Differences between peak lumbar flexion and EMG-off angles (DIFF). Results are given for CONTRA (C) and IPSI (I) movements among participants with LLA, and LEFT (L) and RIGHT (R) movements among controls. Error bars indicate standard deviations, and the asterisk indicates a significant bilateral difference.

Peak lumbar flexion and EMG-off angles decreased with increasing transverse rotation, and this effect likely resulted from additional frontal plane motion and a relatively earlier increase in passive tension developed in contralateral tissues (Ning et al., 2011). Movements here were performed bilaterally, allowing for separate (left and right) evaluations of muscle and tissue contributions to forward-flexion movements. Asymmetric differences (albeit not significant in all conditions) in DIFF angles were observed among both groups, though these asymmetries were more substantial among participants with LLA with increased transverse rotation (Fig. 4). While EMG-off angles were bilaterally similar in the LLA group, peak lumbar flexion angles were significantly smaller during CONTRA movements. Therefore, the ES muscles were activated throughout a longer portion of the range of motion during CONTRA movements. This was reflected by a significantly smaller DIFF angle during CONTRA movements at the 30° transverse rotation, and supports prior evidence of smaller relative passive trunk stiffness in relaxed upright standing during perturbations involving spinal tissues and muscles ipsilateral vs. contralateral to the side of amputation (Hendershot et al., 2013).

Static end range-of-motion (ROM) can be used as an indicator of joint flexibility/stiffness (McHugh et al., 1998). Based on previous evidence of lower trunk stiffness during perturbations involving spinal tissues and muscles ipsilateral to the side of amputation

(Hendershot et al., in 2013), it was expected that peak lumbar flexion angles would be larger overall among persons with LLA and larger during CONTRA movements (i.e., trunk movements directed away from the side of the amputation, and thus involving tissues and muscles on the side of amputation). However, the observed peak lumbar flexion angles were contradictory to these expectations, and may have resulted from contributions of different tissues near the end of trunk ROM. Further, static trunk flexion ROM and the onset of trunk muscle relaxation can be influenced by the strength and relative length of muscles of the trunk and hip, as well as the coordination of trunk and hip movements (Gupta, 2001; Pal et al., 2007). Alterations in muscle properties and recruitment strategies within the lower-limb could therefore influence peak lumbar flexion angles and muscle coordination/timing. Thus, inherent bilateral differences in lower-limb musculature (e.g., tightness due to immobilization or contracture) among participants with LLA could have limited movement near the end of trunk ROM. Such differences in hip muscles between groups may explain the smaller peak lumbar flexion angles observed during CONTRA movements among participants with LLA. Performing the flexion–extension movements freely (i.e., without a pelvic restraint) may provide more insight into the coordination of the lumbar spine and pelvis ("lumbopelvic rhythm") among persons with LLA (Sarti et al., 2001; Dubois et al., 2011). Future work should also investigate dynamic measures of trunk flexibility and ROM.

All participants with LLA in the current study exhibited bilateral relaxation of the ES during sagittally-symmetric movements, and contralateral relaxation during sagitally-asymmetric movements. In contrast, patients with LBP typically do not exhibit a relaxation of the superficial back muscles during flexion movements (Shirado et al., 1995; Watson et al., 1997). An absence of muscle relaxation during forward flexion among persons with current LBP may act to limit motion near the end of the ROM, or may be volitional (guarding) and related to fear of further pain (Geisser et al., 2004; Triano and Schultz, 1987; Trost et al., 2012). However, none of the current participants (in either group) were experiencing LBP at the time of testing. It is also presently unclear how other trunk muscles contributed to the measured responses during these trunk movements. Deep trunk (stabilizing) muscles are more active towards the latter part of forward trunk flexion despite the relaxation of the superficial posterior muscles (Andersson et al., 1996), and can influence joint ROM and stability (Quint et al., 1998). Future work should investigate such muscular contributions to flexion movements among persons with LLA, as well as compare these responses among persons with LLA who are experiencing current (or have a history of) LBP.

In summary, the current study identified some alterations in the synergy between active and passive trunk tissue responses with LLA during trunk movements. Movements involving trunk flexion are frequent among activities of daily living, and alterations in this synergy may lead to an increased risk for spinal instability and LBP during such movements. For persons with LLA, alterations in trunk motion during gait and movement subsequent to LLA may alter the passive contributions to joint stability during trunk movement, thus requiring compensations from active trunk muscle responses. Increases in trunk muscle activities can increase the magnitude of spinal load (Callaghan et al., 1998; Granata and Marras, 1995), and altered trunk neuromuscular control and increases in spinal load have been related to LBP (Adams et al., 2006; McGill 2007; White and Panjabi, 1990). Increased and asymmetric trunk muscle activity could also impact the type of spine loading (i.e., compression vs. shear), particularly during asymmetric trunk movements since asymmetric trunk motions/postures are associated with complex spinal loading (Davis and Marras, 2000). Large muscle forces also increase the loads in passive structures (e.g., disc or ligament), and can lead to errors in proprioceptive feedback (Panjabi, 1992; Solomonow, 2009). Hence, altered/asymmetric synergies between passive and active trunk tissue responses among persons with LLA during trunk movements may increase the risk for low-back injury in this population.

Conflict of Interest

None.

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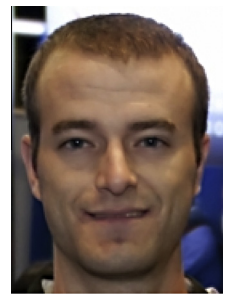

Brad D. Hendershot received a BS (2008) in Bioengineering from Penn State and a PhD (2012) in Biomedical Engineering from Virginia Tech. He is currently a Research Biomechanist with the Henry M. Jackson Foundation, stationed at Walter Reed National Military Medical Center in Bethesda, Maryland. His research focuses on the mechanics and neuromuscular control of human movement, with a particular emphasis on the trunk/spine, and in both healthy and patient populations.

Maury A. Nussbaum received an MS (1989) in Bioengineering and a PhD (1994) in Industrial and Operations Engineering from The University of Michigan. He is currently HG Prillaman Professor of Industrial and Systems Engineering at Virginia Tech in Blacksburg, Virginia. His research interests are (in no particular order) occupational biomechanics and ergonomics, localized muscle fatigue, aging, and balance/postural control.